Exercises

Algebraic problems

Question 8.1: Using *integration by substitution*, find the indefinite integral of the following:

(a) $\int 4\,(5x+6)^3\,dx$

(b) $\int (x-2)^3\,dx$

(c) $\int (x+5)^4\,dx$

(d) $\int (1-2x)^7\,dx$

Question 8.2: Using *integration by substitution*, find the definite integral of the following:

(a) $\int_0^1 4\,(5x+6)^3\,dx$

(b) $\int_0^1 (x-2)^3\,dx$

(c) $\int_0^1 (x+5)^4\,dx$

(d) $\int_{-1}^1 (1-2x)^7\,dx$

Question 8.3: Using *integration by substitution*, find the indefinite integral of the following:

(a) $\int \sin(7x-3)\,dx$

(b) $\int \exp(5x+2)\,dx$

(c) $\int \cos(1-x)\,dx$

(d) $\int \frac{1}{7x+5}dx$

Question 8.4: Using *integration by substitution*, find the definite integral of the following:

(a) $\int_0^{\pi/2} \sin(7x-3)\,dx$

(b) $\int_0^1 \exp(5x+2)\,dx$

(c) $\int_{-\pi/4}^{\pi/4} \cos(1-x)\,dx$

(d) $\int_0^1 \frac{1}{7x+5}dx$

Question 8.5: Using *integration by substitution*, find the definite integral of the following:

$$\int_{-1}^{1} \frac{4}{(2x+1)^3}$$

Question 8.6: We can see that from the table of basic integration laws that

$$\frac{dy}{dx} = x^n \implies y = \frac{x^{n+1}}{n+1}$$

Use integration by substitution to find a general expression for y when

$$\frac{dy}{dx} = (ax+b)^n$$

where a, b and n are constants.

Real-world problems

Question 8.7: The density of air changes with pressure and temperature (because of the ideal gas law). Density is equal to $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1}K^{-1} is the gas constant for air and gravity is equal to $g = 9.8$.. So the differential equation becomes:

$$\frac{dP}{dz} = -\frac{P}{R_a T} g$$

We saw that in Chapter 6 that the solution to this problem is:

$$P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right)$$

However, for simplicity, we assumed that temperature was constant which isn't a very good assumption. Temperature T is actually a function of height in the atmosphere: $T(z) = T_{surface} - 0.01 \times z$, where $T_{surface}$ is the surface temperature. So the final differential equation to solve can be written:

$$\frac{dP}{dz} = -\frac{P}{R_a \left(T_{surface} - 0.01z\right)} g$$

During a day in the summer, the surface temperature at sea level over Tenerife is 290 K and the surface pressure is 100000 Pa. Clouds are observed to form at a height of 1000 m. What are the pressure and temperature at this height? HINT: Rearrange this ODE and use integration by substitution.

Question 8.8: The density of air changes with pressure and temperature (because of the ideal gas law). Density is equal to $\rho = \frac{P}{R_a T}$, where $R_a = 287$ J kg^{-1}K^{-1} is the gas constant for air and gravity is equal to $g = 9.8$.Use integration by substitution to show that the general solution of the hydrostatic equation:

$$\frac{dP}{dz} = -\frac{P}{R_a \left(T_{surface} - 0.01z\right)} g$$

is of the form

$$P(z) = P_0 \exp\left(\frac{100 \times 9.8}{287} \ln\left(\frac{T_{surface} - 0.01z}{T_{surface}}\right)\right)$$

where P_0 is the pressure at $z = 0$.

Question 8.9: In the previous Chapter, we saw the drag force on a sphere which is moving through air is approximately equal to

$$\pi r^2 \frac{\rho}{2} v^2$$

where r is the sphere's radius, v is the velocity it is moving through the air and $\rho = 1.2$ kg m^{-3} is the density of air. A sphere, radius 0.5 m, and mass $m = 1$ kg, is pushed along a

horizontal plane at an initial velocity of $v = 10$ m s^{-1}. However, this time the wind speed is $10m/s$ in the opposite direction of the motion of the sphere. We need to re-write out the above equation slightly to account for the wind speed. If we let w be the wind speed, then the drag force

$$\pi r^2 \frac{\rho}{2}(v+w)^2$$

How far does the sphere move before slowing down to 0.01 m s^{-1}? HINT: You will need to use the appropriate version of Newtons second law and then use integrate by substitution to do this.